

$\therefore$ new $y = (\sqrt{2} - \frac{1}{2})$ is irrational.

$$x^8 = (2(\sqrt{2} + \frac{1}{2}))(\sqrt{2} - \frac{1}{2}) = 2(\sqrt{2} + \frac{1}{2})(\sqrt{2} - \frac{1}{2}) = 2^{2 - \frac{1}{4}} = 2^{7/4} = 2 \cdot 2^{3/4} = 2 \cdot (8^{1/4})^2$$

$$= 2 \cdot \sqrt{2\sqrt{2}}$$

By Defn. 2,
Let, x^8 be rational. $\therefore \exists$ integers a and b in lowest terms, $b \neq 0$
s.t. $2\sqrt{2\sqrt{2}} = \frac{a}{b} \Rightarrow \sqrt{2\sqrt{2}} = \frac{a}{2b} \Rightarrow 2\sqrt{2} = \frac{a^2}{4b^2} \Rightarrow \sqrt{2} = \frac{a^2}{8b^2}$

It's easy to see that a^2 and $8b^2$ are integers. $\therefore b \neq 0, 8b^2 \neq 0$
 $\therefore$ By Defn. 2, $\sqrt{2}$ is rational. This contradicts the fact that $\sqrt{2}$ is irrational from Sec 1.7, Example 11.

$\therefore x^8 = 2 \cdot \sqrt{2\sqrt{2}}$ is irrational $\square$

16. Prove or disprove that if a and b are rational numbers, then a^b is also rational.

Solution: Let, $a = 2$ and $b = \frac{1}{2}$. By Defn. 2, it's easy to see that a and b are rational. $a^b = (2)^{1/2} = \sqrt{2}$. From Sec-1.7, Example 11 we know that $a^b = \sqrt{2}$ is irrational.

$\therefore$ The claim that if a and b are rational numbers, then a^b is rational is False. $\square$

17. Show that each of these statements can be used to express the fact that there is a unique element x such that $P(x)$ is true. [Note that we can also write this statement as $\exists! x P(x)$.]

a) $\exists x \forall y (P(y) \leftrightarrow x = y)$

Solution: By defn. of existential quantifier, there exists some element c_1 in the domain s.t. $\forall y (P(y) \leftrightarrow c_1 = y)$ is True. By defn. of universal quantifier, for any arbitrary element c_2 in the domain, $(P(c_2) \leftrightarrow c_1 = c_2)$ is True.